Education in the Era of ChatGPT: Promise and Pitfalls of Generative AI for Education

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Department of Computer Science and Engineering



Outline

1. ChatGPT/ Generative AI and its promise for education
2. A Critical Look at ChatGPT
3. Educating in the era of ChatGPT/ Advanced AI

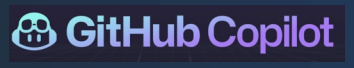
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A brief intro to ChatGPT/ Generative AI and its promise for education

Text Generator



Code Generator



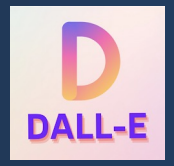


Image Generator



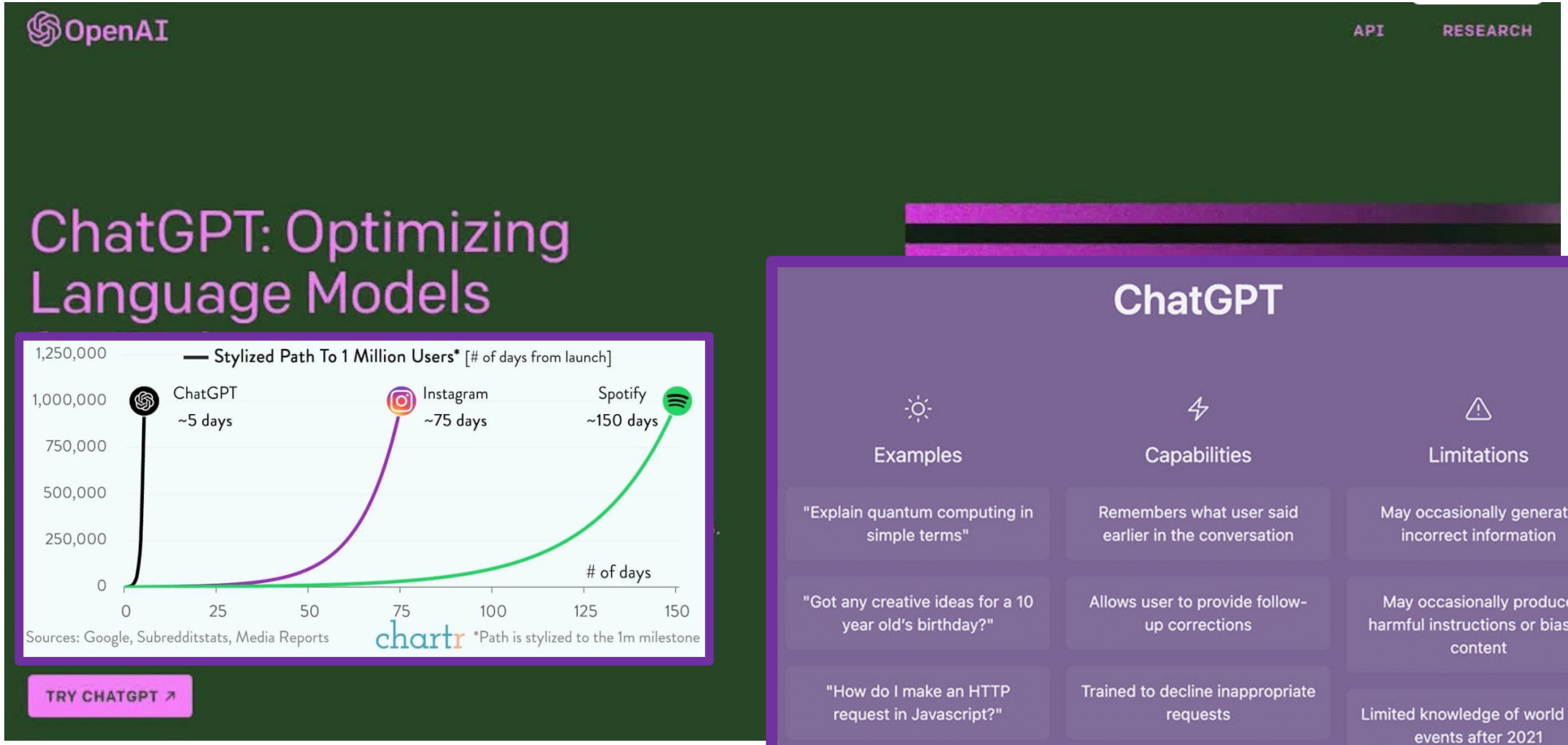
Speech Synthesizer



Conversational AI (ChatBot)

About ChatGPT

https://openai.com/blog/chatgpt/



CODE GENERATION

JU Give me code for the Dijkstra algorithm in Python

Here is the code for the Dijkstra algorithm in Python:

Copy code

```python
from collections import defaultdict
import heapq

def dijkstra(graph, source):
  # Initialize distances and predecessor for each vertex
  distances = defaultdict(lambda: float('inf'))
  predecessor = defaultdict(lambda: None)
  distances[source] = 0

  # Initialize priority queue with all vertices and their distances from the source
  priority_queue = [(0, source)]
  while priority_queue:
    # Extract vertex with minimum distance from the queue
    distance, vertex = heapq.heappop(priority_queue)

    # Update distances and predecessors for all neighbors of the vertex
    for neighbor, weight in graph[vertex]:
      if distances[neighbor] > distance + weight:
        distances[neighbor] = distance + weight
        predecessor[neighbor] = vertex
        heapq.heappush(priority_queue, (distances[neighbor], neighbor))

  return distances, predecessor
```

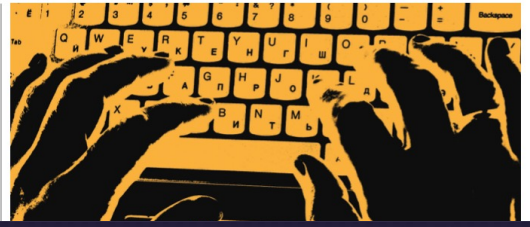
DOI:10.1145/3570220

Matt Welsh

Viewpoint

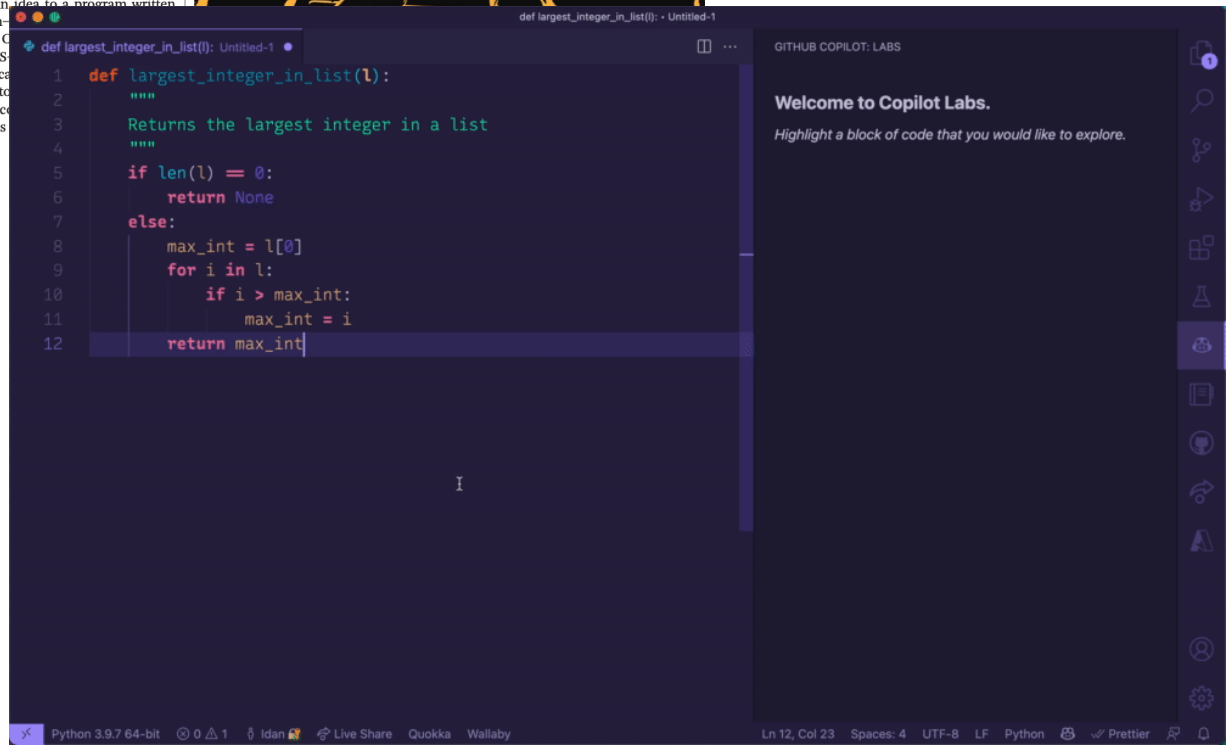
The End of Programming

The end of classical computer science is coming, and most of us are dinosaurs waiting for the meteor to hit.



I CAME OF AGE in the 1980s, programming personal computers such as the Commodore VIC-20 and Apple][e at home. Going on to study computer science (CS) in college and ultimately getting a Ph.D. at Berkeley, the bulk of my professional training was rooted in what I will call "classical" CS: programming, algorithms, data structures, systems, programming languages. In Classical Computer Science, the ultimate goal is to reduce an idea to a program written by a human—like Java or C. Classical CS or sophistica algorithm to tuse Paxos c expressed as

GitHub Copilot



Personalized Learning

The Two-Sigma Effect of Individualized Tutoring



*Students learn best when they receive **formative feedback** about specific qualities of their work along with concrete advice on what they can do to improve.*

VIRTUAL TUTORING & PERSONALIZED LEARNING

JU

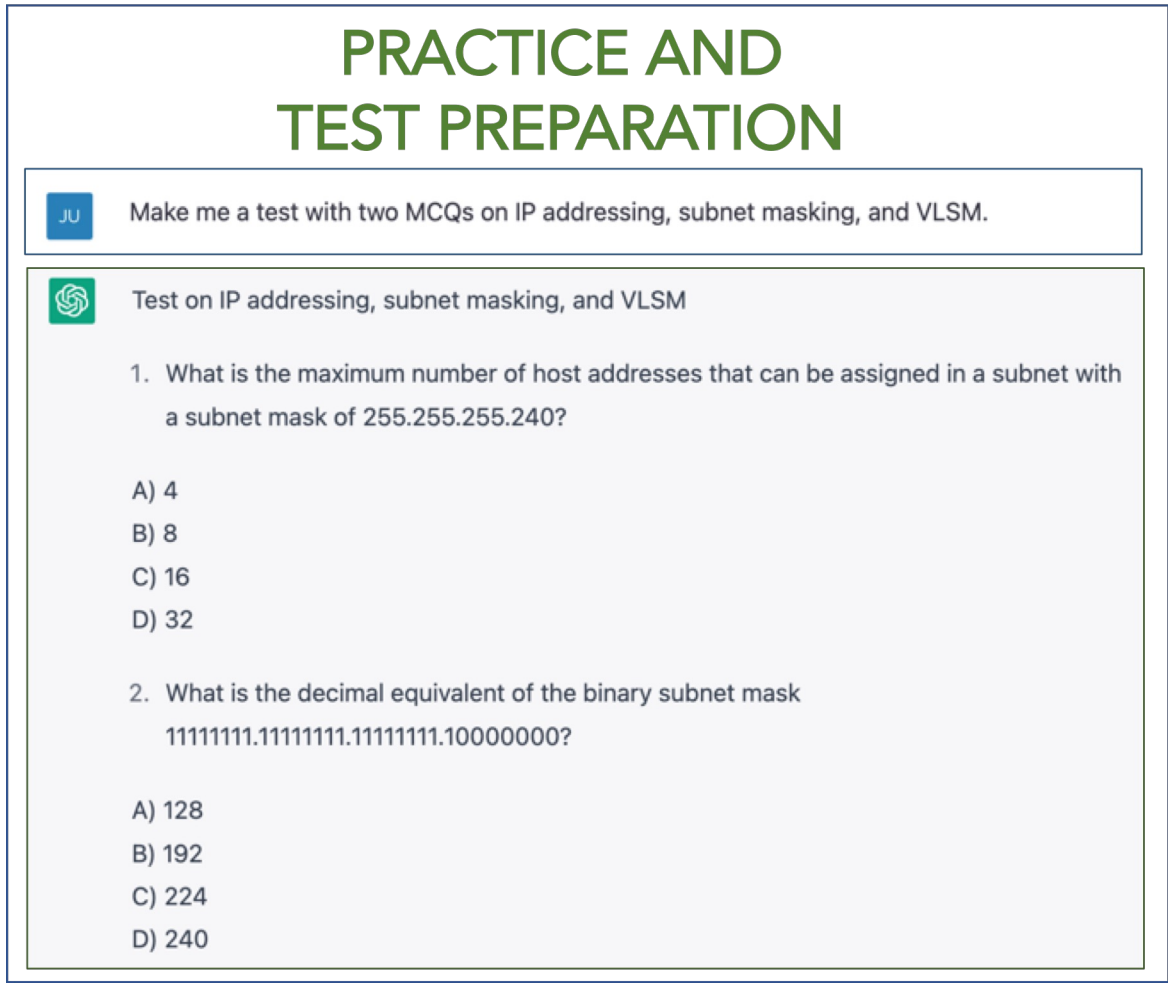
My teacher asked me, "What is the maximum number of host addresses that can be assigned in a subnet with a subnet mask of 255.255.255.240?". My reply was 8 but my teacher said that this is wrong. Why is it wrong?

The maximum number of host addresses that can be assigned in a subnet with a subnet mask of 255.255.255.240 is not 8. The subnet mask 255.255.255.240 allows for 4 bits to be used for host addresses, and each bit can represent 2 values (0 or 1). Therefore, the total number of host addresses that can be assigned in this subnet is 2^4 = 16.

It's important to note that the subnet mask is used to determine the number of host addresses that can be assigned in a subnet. The subnet mask consists of a series of 1s and 0s, and the number of 1s indicates the number of bits that are reserved for the network portion of the IP address. The number of 0s indicates the number of bits that are available for host addresses. By counting the number of 0s in the subnet mask and using the formula 2^n (where n is the number of 0s), you can determine the maximum number of host addresses that can be assigned in a subnet.

Test Preparation

"In virtually all areas of learning, you build better mastery when you use testing as a tool to identify and bring up your areas of weakness."—Brown, Roediger, and McDaniel



Services for Writers/ Researchers/ Essayists

LANGUAGE LEARNING, EDITING, AND WRITING

Edit the following for me: "An important consideration regarding the use of automated thinking and writing tools is that how will it impact the livelihoods of those who are in the business of writing." and provide feedback if it was good writing or if it contained any mistakes or issues.

Here is a revised version of the sentence:

"One important consideration when using automated thinking and writing tools is how they will affect the livelihoods of those who make their living as writers."

This revised version is clearer and more concise than the original sentence. I did not identify any mistakes or issues in the original sentence, but I made a few changes to improve its clarity and flow. Specifically, I replaced the phrase "regarding the use of" with the more concise "when using," and I rephrased the final part of the sentence to be more concise and to use more specific language. Overall, the revised sentence is a well-written and clear expression of the idea that the use of automated tools could have an impact on the livelihoods of writers.

STORY WRITING AND PERSUASIVE WRITING

JU Can you tell me a nice story of how too much use of a technology can actually be detrimental

Sure, here is a story about the negative consequences of overusing a technology:

There was once a small village nestled in the mountains, where the people lived a simple and self-sufficient life. They grew their own food, raised animals, and spent their days working and playing in the great outdoors.

One day, a group of traveling merchants arrived in the village, bringing with them a new technology that promised to revolutionize the way the villagers lived. It was a machine that could do the work of many people, and the merchants convinced the villagers to buy it, promising that it would make their lives easier and more efficient.

At first, the villagers were excited about the new technology. They used it to plow their fields, churn their butter, and even make their meals. But as time passed, they found that they were relying on the machine more and more, and they began to neglect their own skills and abilities. They stopped gardening, stopped caring for their animals, and stopped doing many of the tasks that had once been a part of their daily lives.

As the villagers became more and more dependent on the machine, they started to lose their sense of community and connection with each other. They spent more time indoors, sitting in front of screens and interacting with the machine, and less time outdoors, working and playing together.

One day, the machine broke down, and the villagers realized that they had lost the knowledge and skills they needed to repair it. They were completely reliant on the machine, and without it, they were lost. They had to rely on the merchants to fix it, and they had to pay a high price for the repair.

The villagers learned a valuable lesson that day: that too much reliance on a single technology can be detrimental, and that it is important to maintain their own skills and abilities, and to nurture their sense of community and connection with each other.

Mathematics Questions

Find the derivative of the function f(x) = 3x^2 + 2x - 1. Describe your working

To find the derivative of the function f(x) = 3x^2 + 2x - 1, we need to use the rules of differentiation. The derivative of a function is a measure of how the function is changing at a particular point.

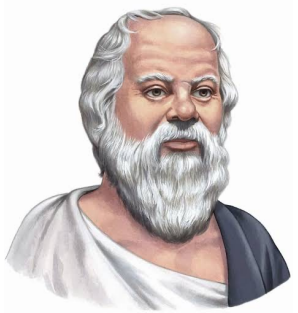
The derivative of a constant term (such as -1 in this case) is always 0, so we can ignore that part of the function. The derivative of a term with a coefficient of 1 (such as 2x in this case) is just the coefficient times the variable (in this case, 2x), so the derivative of 2x is 2.

The derivative of a term with a coefficient other than 1 (such as 3x^2 in this case) is the coefficient times the derivative of the term with the coefficient of 1 (in this case, 3 times x), so the derivative of 3x^2 is 3x.

Now that we have found the derivative of each term in the function, we can add them up to get the derivative of the whole function:

f'(x) = 3x + 2 = 6x + 2

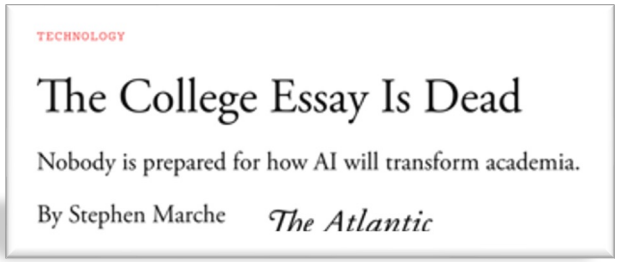
This is the final answer. The derivative of the function f(x) = 3x^2 + 2x - 1 is f'(x) = 6x + 2.



Socratic critique of technology

Such tools may impart "not truth but only the semblance of truth" and may appear to be omniscient and will generally know nothing."

The implications of all this for education/ university


TECHNOLOGY

The College Essay Is Dead

Nobody is prepared for how AI will transform academia.

By Stephen Marche *The Atlantic*

MOTHERBOARD
TECH BY VICE

ChatGPT Is Passing the Tests Required for Medical Licenses and Business Degrees

ChatGPT passes exams from law and business schools


By Samantha Murphy Kelly, CNN Business
Updated 1:35 PM EST, Thu January 26, 2023

ChatGPT passes Wharton Business School test: research paper

Engineering Education in the Era of ChatGPT: Promise and Pitfalls of Generative AI for Education

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IEEE EDUCON 2023
IEEE Global Engineering Education Conference

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A Critical Look at ChatGPT

Ledger of Harms

Opaqueness

Unpredictability

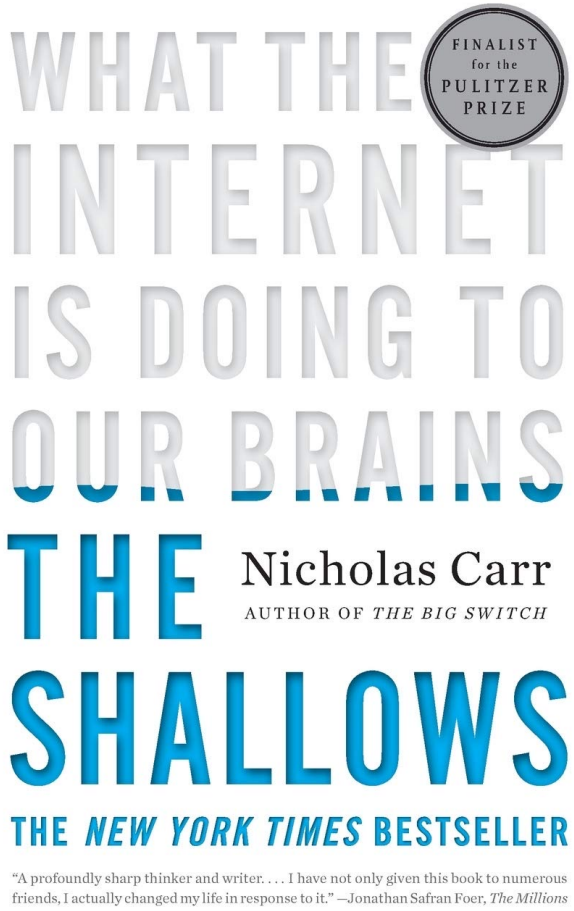
Hallucinations

Privacy violation

Harmful output (bias, mis/disinfo)

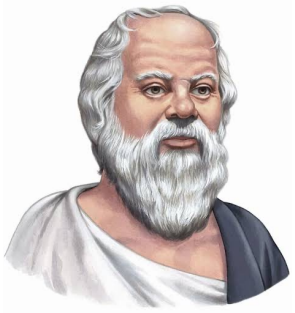
Inefficiency

The risk of outsourcing thinking to technology



Plato's Argument Against Writing

"If men learn this, it will implant forgetfulness in their souls. ***They will cease to exercise memory because they rely on that which is written, calling things to remembrance no longer from within themselves, but by means of external marks.***"



Such tools impart "not truth but only the semblance of truth" and that those who adopt it will appear to omniscient and will generally know nothing."

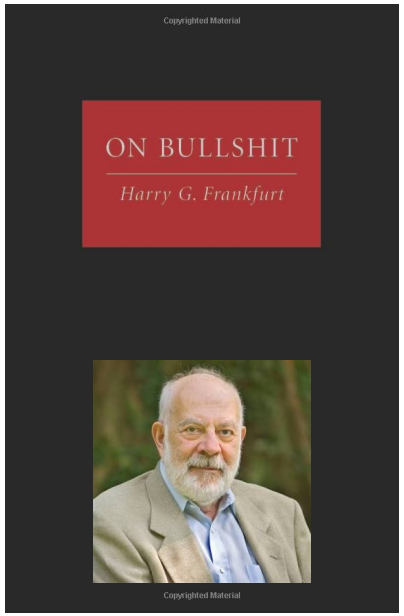
Is ChatGPT reliable and trustworthy?

ChatGPT is "high-tech plagiarism" for lazy learners, says Noam Chomsky





ChatGPT in less flattering terms



ChatGPT is a bullshit generator. But it can still be amazingly useful



Arvind Narayanan and Sayash Kapoor
Dec 6, 2022

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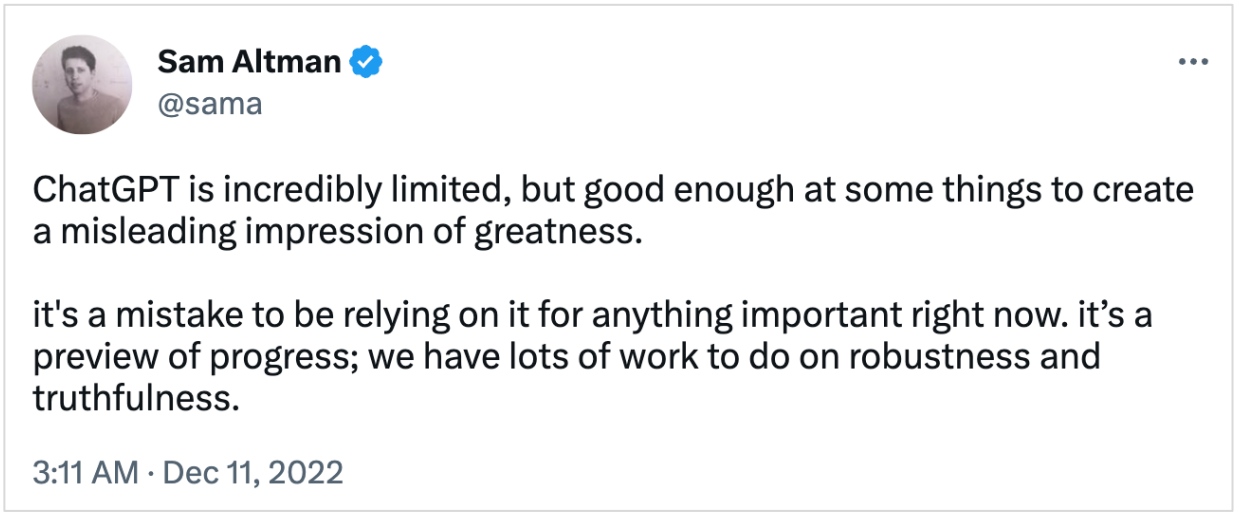
The philosopher Harry Frankfurt defined bullshit as speech that is intended to persuade without regard for the truth.

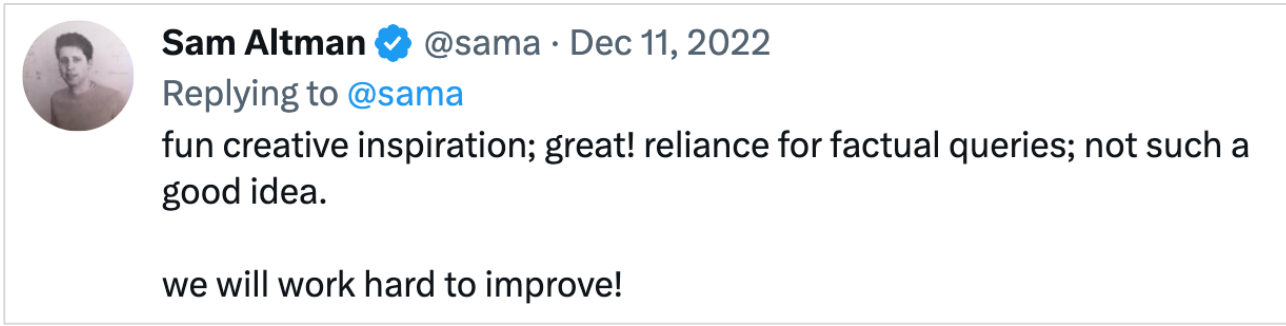
Caveat Emptor: Buyer Be Aware

Let's say it's highly accurate (e.g., 90% of the time, it is factual).

Can we use it for critical tasks?

***The problem is that we do not know which 90%**, and worse, ChatGPT doesn't know either!"*





Yann Le Cun (FB) on Large Language Models (LLMs)



Yann LeCun
@ylecun

My unwavering opinion on current (auto-regressive) LLMs
1. They are useful as writing aids.
2. They are "reactive" & don't plan nor reason.
3. They make stuff up or retrieve stuff approximately.
4. That can be mitigated but not fixed by human feedback.
5. Better systems will come

4:02 PM · Feb 13, 2023 · **723.8K** Views

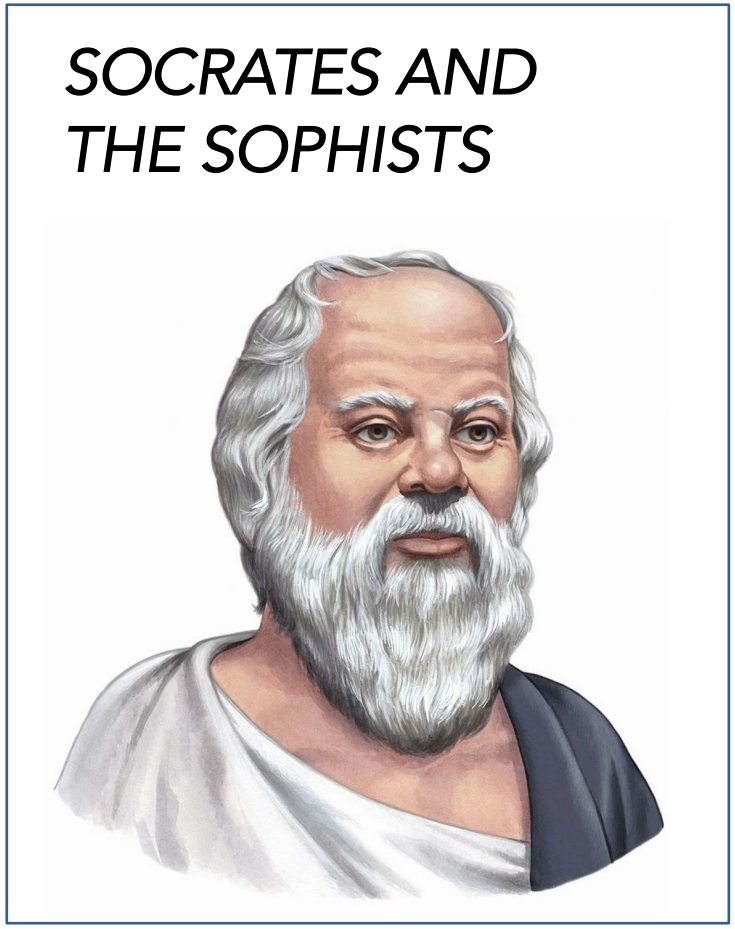
6. Current LLMs should be used as writing aids, not much more.
7. Marrying them with tools such as search engines is highly non trivial.
8. There *will* be better systems that are factual, non toxic, and controllable. They just won't be auto-regressive LLMs.

3

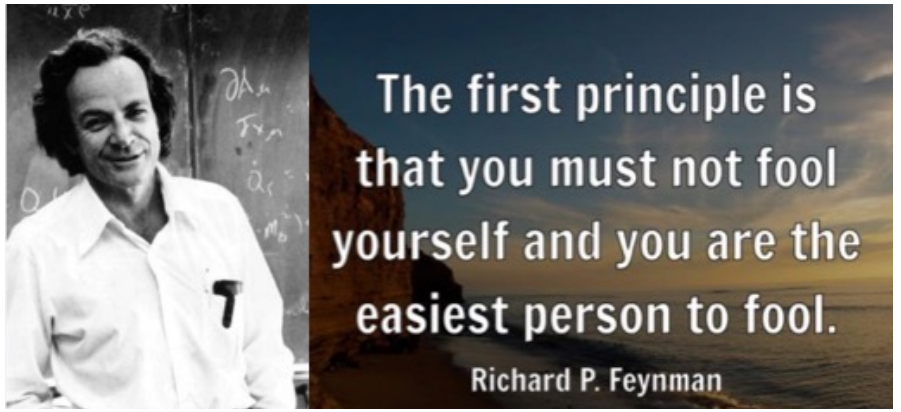
Educating in the era of ChatGPT/ Advanced AI

A brief detour into the philosophy of education

Vignette 1: not getting fooled by superficiality

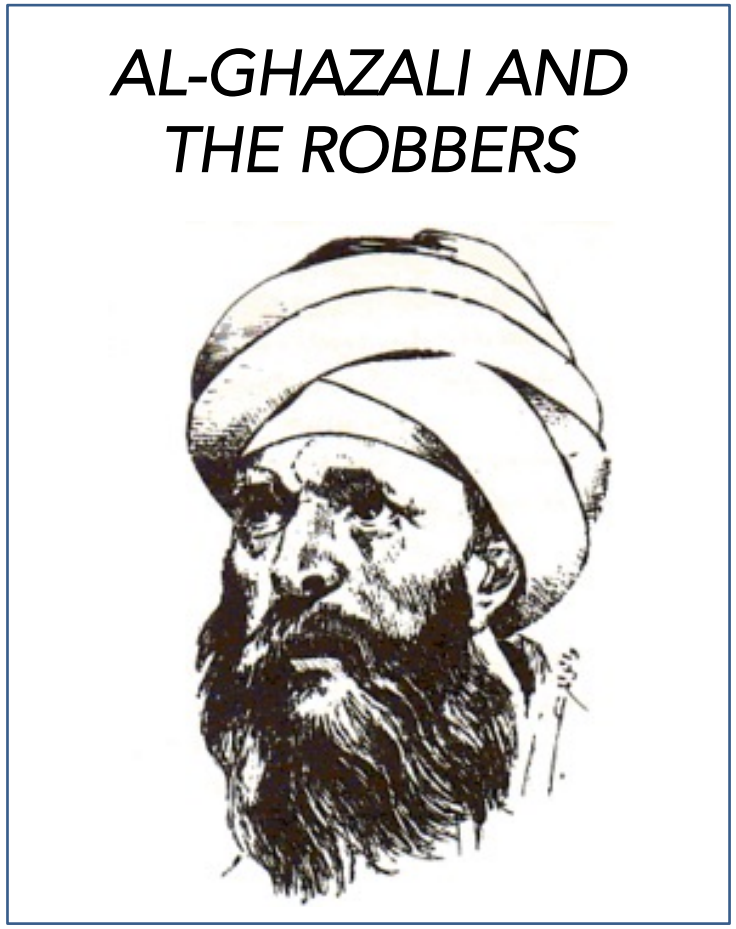


Appearences can be deceiving



Challenge for educators:
how to inculcate intrinsic motivation in students necessary for mastery instead of superficialities.

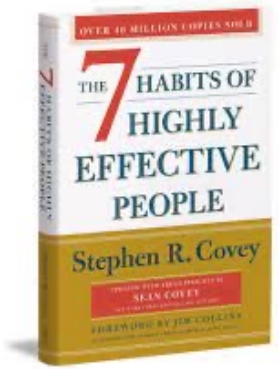
Vignette 2: knowledge must be within



Education and knowledge is inside you

Challenge for educators:
how to use technology so that learners enhance their abilities (cognitive, affective, and psychomotor) in a way that it does not diminish when the technology is not there.

Beware of Shortcuts

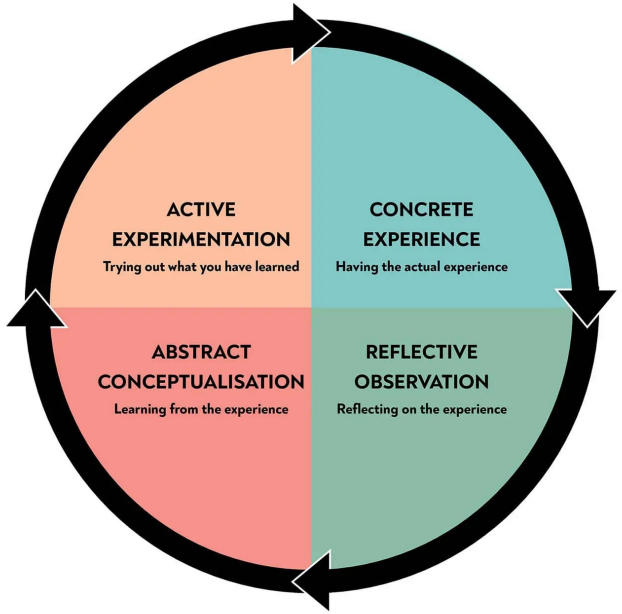




This is why cramming does not work!

(Otherwise, we could give all the answers/info at the start)

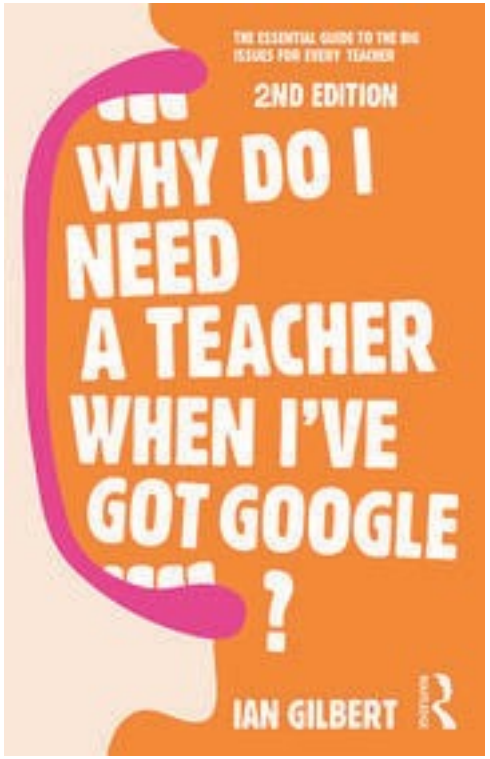
Kolb's Learning Cycle



Paradoxically, some difficulties and failures are desirable for true learning to happen

"An expert is a person who has made all the mistakes that can be made in a very narrow field." - Niels Bohr

Can learning be automatized and outsourced?



You need to train your mind by doing this and go through the process to self-discover or make your concept your own.

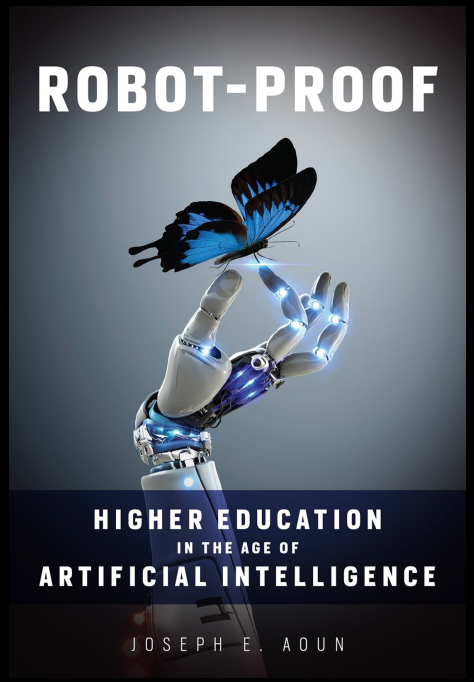
Information Technology and Educational Amnesia

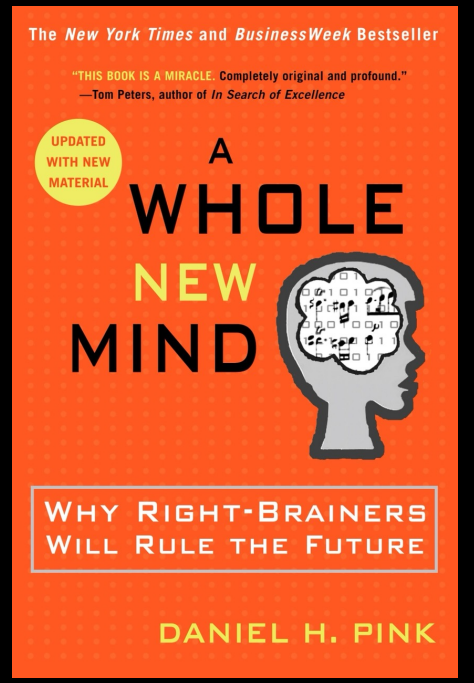
LANGDON WINNER
Rensselaer Polytechnic Institute, Troy, New York, USA

No matter how wonderful the technology (TV, VCR, Internet, Google, ChatGPT), technology alone is not sufficient.Technology can amplify intent and capabilities not substitute for them.

You can outsource labor but not thinking.

Making our students AI Proof

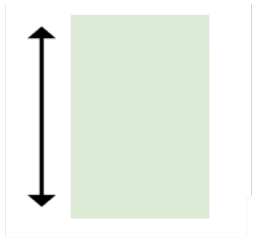




T-shaped engineers and training for breadth + depth

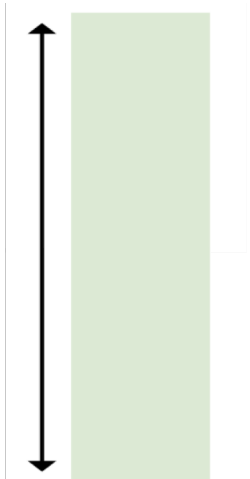


ChatGPT knows everything superficially but understands nothing



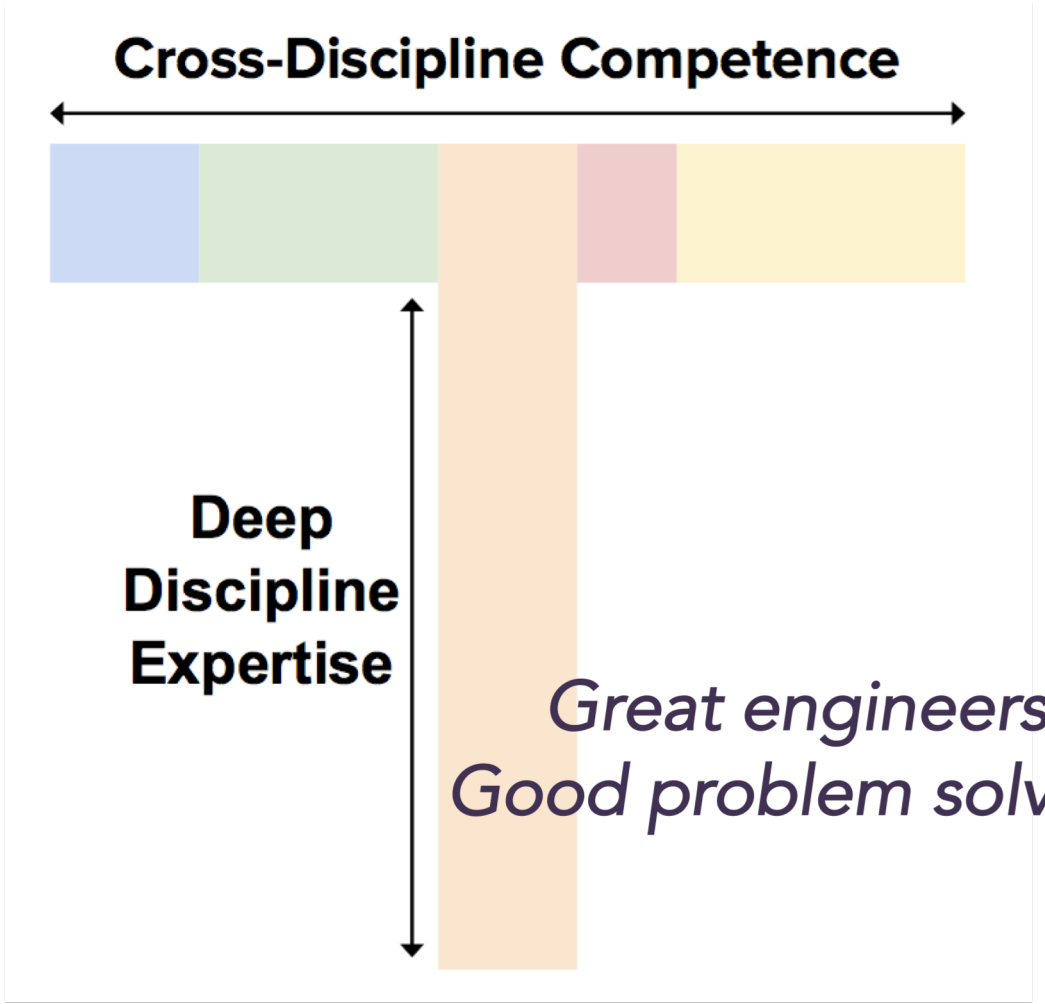
Bad engineers

Superficial narrow knowledge

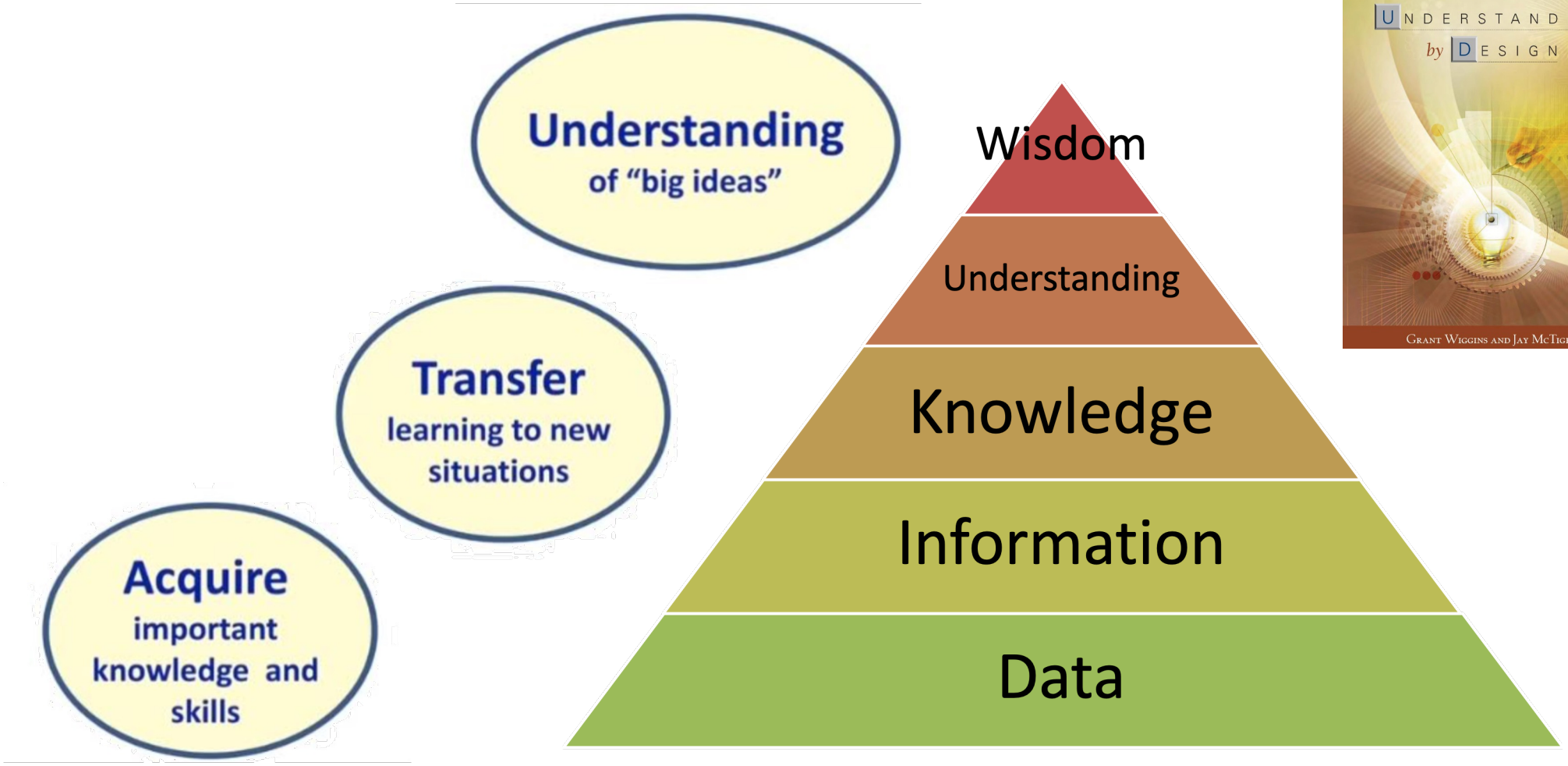


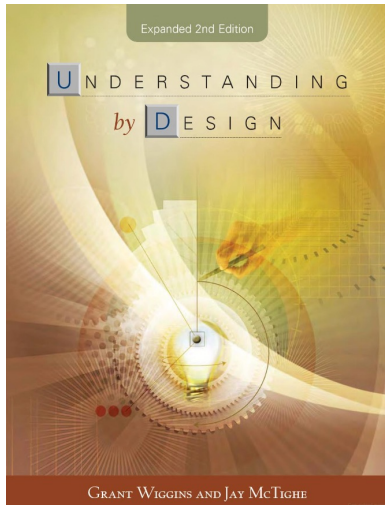
Good engineers
Bad problem solvers

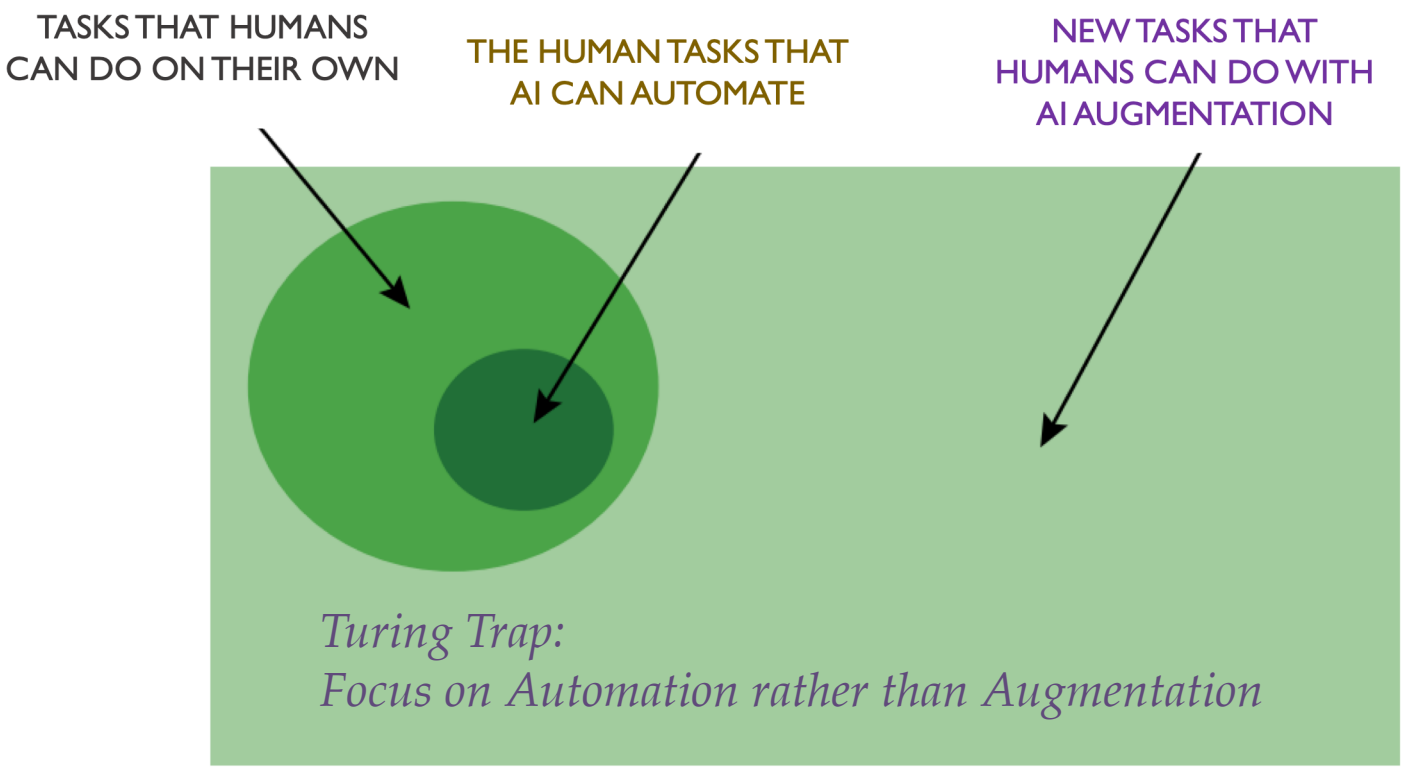
Deep (but narrow) expertise



Focusing on Understanding and Wisdom









AI for Education, especially, should be:

(1) Human-Centered; and

(2) based on Intelligence Augmentation

Scaling feedback with AI to improve student learning

لا يتعلم العلم مستحي ولا مستكبر —

two people do not learn: the ***shy*** and the ***arrogant***

1. A learner should articulate their understanding

2. The understanding is assessed

3. Actionable suggestions on how to improve the understanding needs to be provided.

4. The learner has an improved understanding



The cycle needs to be repeated



Tips and Guidelines



Developing expertise and authenticity in education

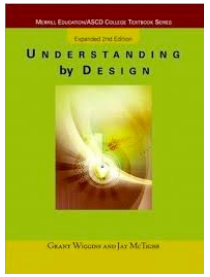


AUTHENTICITY
IN EDUCATION

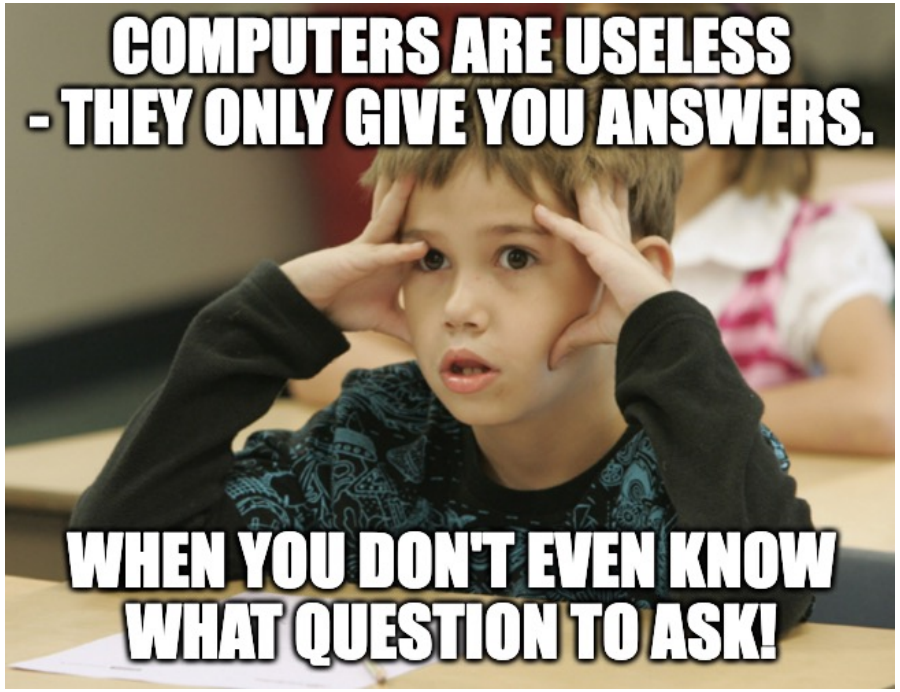


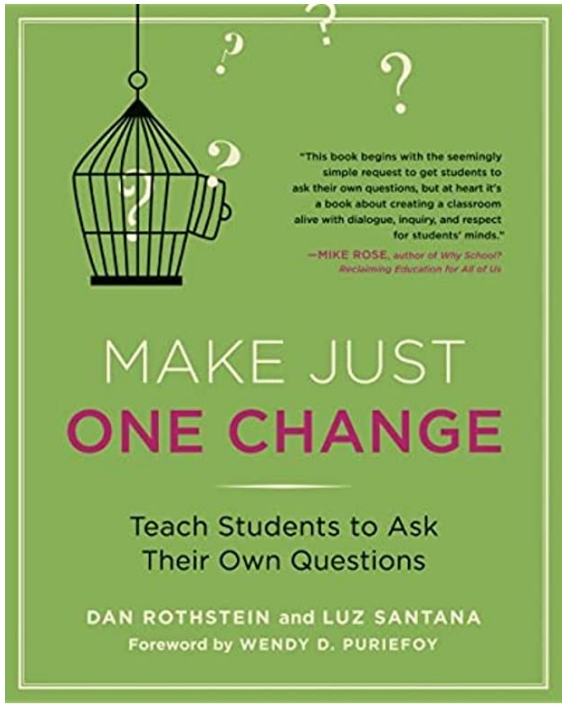
A performance task, problem, or project, is authentic if it:

(1) is realistic;
(2) requires judgment and innovation;
(3) engages the learner in "doing" the subject;
(4) replicates or simulates real-world contexts;
(5) assesses a learner's ability to deal with an open-ended complex task by using a repertoire of knowledge and skills;
(6) does not put artificial constraints on opportunities to practice, consult resources, obtain feedback, and refine performance.



Enhancing IQ (inquiry quotient)

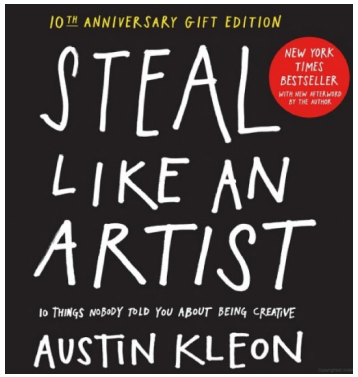


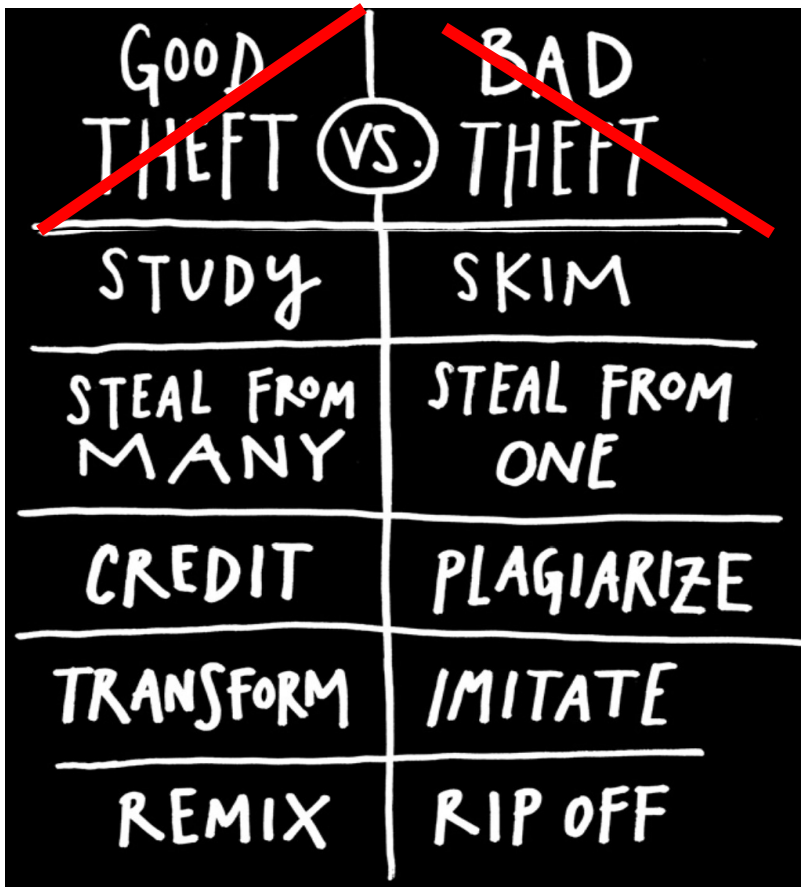


To engage students in meaning making and to develop self-directed learners, students should be encouraged to develop their own questions— and to pursue the answers through active inquiry (i.e., to become researchers)

The importance of value creation

INSPIRATION PLAGIARISM





Talking about an "AI Tools Fair Usage Policy"

<u>Important:</u> Communicate with your students about acceptable AI usage for your courses

You should not steal/ copy without attribution.
You should not deceive.
Cite and credit AI
Cite appropriate sources to support claims in the answer that you get from AI (or be skeptical of the claims)

Guide the students to use AI as a tool to learn, not just to produce content.

"Laws" of Human Intelligence Augmentation

[Aimed at Learners]

First Law 

When the AI is readily accessible, your work should be no worse than that produced by the AI alone.
(Embrace change and leverage AI)

Second Law 

(Corollary of Law 1)
When AI is there, and you don't want to be made irrelevant, maintain your competitive advantage
(Become **AI Proof**)

Third Law 

After becoming users of AI, don't let your inherent unaided capabilities drop
(**Don't outsource your thinking muscle allowing it to atrophy**; i.e., do no harm [*a la* Asimov; Hippocrates]

Principles of Human Intelligence Augmentation

[Educator's Edition]



Principle 1

The learning process should be centered around the student, rather than on the teacher, curriculum, or AI

(Be learner-centric and make your education human-centered)



Principle 2

AI should be utilized in a way that facilitates scalable iterative rapid formative feedback

(Scale formative feedback using AI)



Principle 3

Teach the students the critical thinking skills necessary for judging and correcting the output of AI models

(Equip students with AI error-correction abilities.]

Thank you!

Your comments and input are most welcome

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